

The Void Is Not False: Recovering the Unmarked State in Logic from the Calculus of Indications

Rowan Brad Quni-Gudzinis

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Abstract

Boolean logic treats `false` as a primitive truth value, and classical set theory treats the empty set as a primitive object. This paper argues that both commit a foundational category error: they reify the void — the unmarked state before any distinction — into a member of the system. In the calculus of indications, the unmarked state is not a second value; it is the absence of the mark, the background against which every distinction is drawn. `false` is itself a mark, a positive distinction, not the void. The paper develops this critique as its defensible claim, and then outlines a reconstruction program in which negation is the act of crossing a boundary, excluded middle is a local property of clearly drawn distinctions, and re-entry produces an oscillating state that is neither true nor false. The reconstruction is labeled conjecture: its central claims are unproven, and the evidential-weight boundary between the critique and the program is stated explicitly. Claims marked [my conjecture] carry falsifiability conditions and are not presented as established results.

Keywords: void; false; calculus of indications; laws of form; Boolean logic; unmarked state; negation

1. Introduction

Classical logic begins with two truth values, true and false, and classical set theory begins with the empty set. This paper examines a shared presupposition of both: that absence can be a member of a system. In the calculus of indications, the primitive is the mark, and the unmarked state is not a second object but the absence of the mark [1, 2]. `false`, by contrast, is a mark: the boundary that separates the inside from the outside may be oriented so that the outside is called false, but the orientation is an act, not a primitive [1].

The critique developed in Sections 2–4 is that Boolean logic reifies the unmarked state into the value `false`, and set theory reifies it into the empty set. Both treat absence as a member of the system. The critique is the paper’s defensible claim.

Section 5 outlines the reconstruction program — a logic of distinctions with inside, outside, and oscillating states — and labels its central conjectures with falsifiability conditions. Section 6 shows that classical Boolean logic is the static restriction of this logic. Section 7 states the epistemic boundary of the work explicitly.

2. The Unmarked State and Its Reification

The calculus of indications begins with the unmarked state: no boundary has been drawn, no distinction made [1]. The first act is the drawing of a distinction, which creates three things at once: a marked state (the inside), an unmarked state (the outside), and the boundary between them [1].

The formal content of this section is established [1, 2]. The interpretive claim — that the unmarked state is a background, not a value — is the thesis of this paper.

Boolean algebra writes two values, **0** and **1**, and identifies **0** with false. Set theory writes the empty set $\emptyset$ and treats it as an object. The critique: **0** and $\emptyset$ are *marks* (positive distinctions), not the unmarked state. The empty set is a set; false is a truth value; both are things. The void is not a thing. [my conjecture – the interpretive claim] This claim is disconfirmed if a formal account is produced in which the unmarked state functions as a member of a system without any marking operation — i.e., absence is a value prior to any act of indication. No such account is known.

3. Laws of Form as a Logic of Crossing

The mark is a boundary with an inside and an outside. Negation is the act of crossing that boundary, not the mapping to a second primitive value [1, 2]. Truth and falsehood are orientations of the boundary, relative to a drawn distinction.

The law of crossing: a boundary crossed twice returns to the unmarked,

$$\overline{\overline{x}} = x.$$

The law of calling: a mark indicated twice is the same mark indicated once,

$$AA = A.$$

These laws are established [1, 2]. The interpretive claim that negation is an act rather than a complement is [established – Spencer-Brown 1969] within the calculus; its bearing on Boolean logic is the critique of the next section.

4. The Category Error (the Critique)

Boolean logic commits a category error when it treats false as the absence of truth. false is a mark; it is the orientation of a boundary that selects one side. The unmarked state is not the negation of true; it is the condition under which no boundary has been drawn, and hence no truth value exists.

Set theory commits the parallel error with the empty set. The empty set $\emptyset$ is a set — it has an identity, a cardinality (zero), and a place in the cumulative hierarchy. The void is not a set, not a proposition, not a number. It is the precondition for any of these. [my conjecture – the interpretive claim] Disconfirmed if the empty set can be shown to function as the unmarked state rather than as a marked object.

The logical consequence: excluded middle,

$$P \vee \neg P = \text{true},$$

is not a cosmic law. It presumes that the distinction P has already been drawn cleanly. Before the mark, there is no P , no $\neg P$, and no truth value. Excluded middle is a local rule for marks that have already been made. [my conjecture] Disconfirmed if a formulation of the calculus of indications is produced in which excluded middle holds before any boundary is drawn.

This critique connects directly to the existing literature on non-classical logic. Paraconsistent logic rejects explosion while keeping static valuations [3, 4, 5]; intuitionistic logic rejects excluded middle while keeping all propositions stable [6]; trivalent logics add a third static value [7]. None of these has the unmarked state as background and an oscillating state as a dynamic value. The distinction of the present critique is the *category error itself*: false is a mark, the void is not a value. [my conjecture]

5. The Reconstruction Program: A Logic of Distinctions

This section outlines the research program that would turn the critique of Section 4 into a constructive logic. **The claims in this section are [my conjecture] unless stated otherwise. They are not established results and are not presented as evidence; the evidential-weight gate applied in Section 7 blocks them.**

5.1 States as configurations of marks

A *state* is a finite configuration of marks. The void is the empty configuration — not a member of the value set, but the background against which configurations exist. Truth values are not primitives; they are orientations of boundaries.

5.2 The oscillating state

[my conjecture] **Conjecture V1:** the re-entrant equation $f = \bar{f}$ (a mark indicating its own negation) has no static solution but has a *dynamic* solution: the configuration oscillates between marked and unmarked. This “imaginary truth value” (Spencer-Brown’s Chapter 11) is neither true nor false; it is a phase.

Conjecture V1 is disconfirmed if the re-entrant equation has a static solution in the calculus, or if the oscillation is reducible to a static third value.

5.3 The discrimination test

[my conjecture] **Conjecture V2:** the semantics of distinctions (inside / outside / oscillating) is genuinely different from: - **intuitionistic logic** — which rejects excluded middle but keeps all propositions as stable truths; it has no oscillating state; - **paraconsistent logic** — which rejects explosion but keeps static valuations; it has no unmarked-state-as-background; - **trivalent (neutrosophic) logic** — which adds a third *static* value; the distinction-logic's third state is *dynamic* [7].

The confirmation-seeking test (locked in the project plan): exhibit a formula whose semantic value in distinction-logic is oscillation while intuitionistic and paraconsistent semantics assign a static value. If no such formula exists, the distinction-logic semantics is a re-labeling — disconfirmed. This is the KIF-60 boundary condition: until the discrimination formula is exhibited, the reconstruction carries zero evidential weight as a new logic.

5.4 Classical fragment recovery

Classical Boolean logic is the *restriction of distinction-logic to static configurations* (no re-entry, no oscillation). Excluded middle holds locally for clearly drawn boundaries (every configuration is either inside or outside a given boundary), which recovers the valid classical fragment without treating the void as a value. [established – Boundary Algebra literature; 8, 9]

6. Boolean Logic as the Static Restriction

If the reconstruction holds, Boolean logic is not wrong; it is static. It is the fragment of distinction-logic in which no re-entry is permitted and no oscillation occurs. This reading explains why Boolean algebra is adequate for finite, well-behaved systems while failing to capture self-reference, paradox, and phase. [my conjecture]

The formal literature on the correspondence between the calculus of indications and Boolean algebra [8, 9] establishes that the mark calculus can express Boolean algebra. The critique of this paper is not that the correspondence fails, but that the *semantic reading* of the unmarked state as false is the error. [established – Meguire 2003; Bricken 2023]

7. Discussion and Limits

The epistemic boundary of this paper is stated in full:

1. **The critique (Sections 2–4) is the defensible claim.** The law of calling and the law of crossing are established [1]; the observation that false is a mark and the void is not a value is a direct, checkable consequence. The critique does not depend on the reconstruction program.
2. **The reconstruction (Section 5) is a research program.** Conjectures V1 and V2 carry explicit disconfirmation conditions. Until they are proven, they carry zero evidential weight as a new logic: the framework was built to produce oscillation, so producing it is not yet evidence. This is the retrodiction boundary.

3. **The framework has degrees of freedom.** Any structure can, in principle, be re-described as “distinctions plus oscillation.” The specific constraint is the discrimination test: the oscillating state must be shown to differ from both static paraconsistent dialetheia and static trivalent values. The absorption risk — declaring every counterexample a new oscillation mode — is managed by fixing the allowed oscillation modes in advance.
4. **The claims of Section 6 are interpretive re-descriptions of established mathematics** (boundary algebra), not new predictions.

The strategic position of the paper is therefore: the critique stands now; the reconstruction is a program with a clear proof obligation.

8. Conclusion

The void is not false. False is a mark, the orientation of a boundary; the void is the unmarked state from which all boundaries are drawn. Boolean logic’s identification of the two is a category error that produces a static, tree-only logic without re-entry. The critique is the paper’s claim. The reconstruction — a logic of distinctions with inside, outside, and oscillating states — is the paper’s program, labeled conjecture, with falsifiability conditions stated and the retrodiction boundary drawn. The discrimination formula of Conjecture V2, or its refutation, is the next step.

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